Pinecone Weather Machine

A humidity sensor with no battery and no code

At a glance: Build a humidity sensor with no battery and no code. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: humidity response, hygromorphs, passive sensing, biomimicry (Understand).
- Design a controlled test by changing one variable while holding the others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- dry pine cones: 2 to 3 per team, about 8 to 12 total (collect and dry them ahead of time)
- index cards: the moisture-absorbing layer, about 6 to 8 per team, one 100-pack covers the camp
- wood skewers: a few per team for the backing/spine
- spray bottles: 1 per team (4) plus 1 spare, filled with plain tap water
- rulers: 1 per team (4) plus 1 spare, used as the output scale (12 in rulers are fine)
- clear tape (packing or cellophane): shared, 1 roll plus 1 spare
- paper fasteners: a few per team for a pivoting dial pointer
- timer: 1 per team (4) plus 1 spare or staff phones, to time the response

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up 4 stations, one per team, with the materials above, sorted and ready, and designate one spray test spot (over paper towels) where all spraying happens so the floor stays dry.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE is needed: this is a dry tabletop build with one small water spray. Instead, inspect the cones for sap, insects, and sharp scales, set out a few paper towels at the spray spot, and prepare a paper-only bilayer strip for any pine-sensitive student.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Build a humidity sensor with no battery and no code." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record 0:44 - 0:56

At the spray spot, teams wet the paper layer with a fixed number of spray pumps, start the timer, read how far the tip moves on the ruler and the time to reach its farthest point, then let it dry and confirm it returns. Log this baseline (distance moved and response time) before changing anything.

Phase 6 Redesign 0:56 - 1:08

Each team changes exactly one variable (strip length, paper-layer thickness, OR spray pumps held constant), predicts the effect, and re-tests with the same spray-and-dry procedure to compare distance moved and response time against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wipe the spray spot dry, return the cones, spray bottle, ruler, skewers, and tape to the bin, and recycle used card strips; there is no PPE to remove.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the spray-and-dry test: the same number of spray pumps, the same technique, and the same start position, then repeat it before recording.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to hygromorphs.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	100

CLEANUP AND SOURCES

Return the cones, spray bottles, rulers, skewers, and tape to the bin, recycle used index-card strips, wipe the spray spot and any wet surface dry, and collect score slips.

SOURCES AskNature Pine Cones Strategy